

## EXPERIMENT – 6

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### **Snort IDS/IPS to Detect and Prevent Real-Time Threats Using TryHackMe Platform**

#### **Aim**

To understand and demonstrate the working of Snort as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by detecting real-time network threats using the TryHackMe platform.

#### **Tools / Platform Required**

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Snort IDS/IPS Tool (Pre-installed in AttackBox)

#### **Theory (Brief)**

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are essential components of network security used to monitor, detect, and prevent malicious activities. Snort is an open-source, signature-based IDS/IPS capable of real-time traffic analysis and packet inspection.

TryHackMe provides a safe and legal environment to practice IDS/IPS concepts by simulating real-world attacks and defenses without impacting live systems.

#### **Steps to Perform the Experiment**

##### **Step 1: Login to TryHackMe**

1. Open browser and go to: <https://tryhackme.com>
2. Login using registered credentials.

##### **Step 2: Open Snort Room**

1. In the search bar, type **Snort**.
2. Select a room related to **Snort IDS/IPS**.
3. Click **Join Room**.

##### **Step 3: Start the AttackBox**

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

##### **Step 4: Verify Snort Installation**

1. In the terminal, check Snort version:

2. `snort --version`
3. Ensure Snort is successfully installed.

### Step 5: Run Snort in IDS Mode

1. Start Snort in console mode:
2. `sudo snort -A console -q -c /etc/snort/snort.conf -i eth0`
3. Snort begins monitoring network traffic in real time.

### Step 6: Detect Network Threats

1. Perform network scanning or simulated attacks as instructed in the TryHackMe room.
2. Observe alerts generated by Snort.
3. Analyze alerts indicating suspicious or malicious activity.

### Step 7: Configure Snort as IPS (Optional)

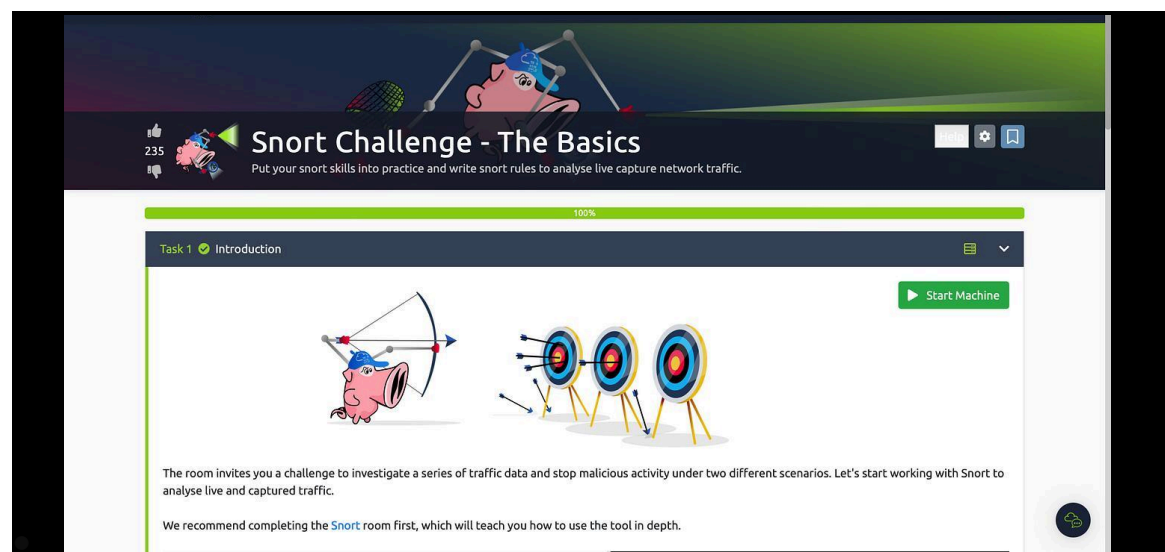
1. Run Snort in inline (IPS) mode:
2. `sudo snort -Q --daq afpacket -c /etc/snort/snort.conf -i eth0`
3. Configure rules to drop malicious packets.
4. Observe blocked traffic in real time.

### Step 8: Analyze Alerts

1. Review generated alerts on the terminal.
2. Identify attack types such as port scanning, brute force, or suspicious packets.

### Output

- Snort successfully detects malicious traffic.
- Real-time alerts are displayed in the terminal.
- Screenshots of alerts and completed tasks are recorded.



Ubuntu22.04 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Apr 15 00:00

ti@ti-VirtualBox: ~

```
o" )~ Version 2.9.15.1 GRE (Build 15125)
' ' By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
    Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.

    Copyright (C) 1998-2013 Sourcefire, Inc., et al.
    Using libpcap version 1.10.1 (with TPACKET_V3)
    Using PCRE version: 8.39 2016-06-14
    Using ZLIB version: 1.2.11

    Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.1 <Build 1>
    Preprocessor Object: SF_POP Version 1.0 <Build 1>
    Preprocessor Object: SF_DNS Version 1.1 <Build 4>
    Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
    Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
    Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
    Preprocessor Object: appid Version 1.1 <Build 5>
    Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
    Preprocessor Object: SF_SDF Version 1.1 <Build 1>
    Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
    Preprocessor Object: SF_GTP Version 1.1 <Build 1>
    Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
    Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
    Preprocessor Object: SF_SIP Version 1.1 <Build 1>
    Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
    Preprocessor Object: SF_SSH Version 1.1 <Build 3>

    Snort successfully validated the configuration!
    Snort exiting
ti@ti-VirtualBox:~$
```

jay@Ubuntu: ~

```
jay@Ubuntu:~$ sudo snort -q -l /var/log/snort -i enp0s3 -A console -c /etc/snort/snort.conf
[sudo] password for jay:
^C*** Caught Int-Signal
jay@Ubuntu:~$ sudo snort -q -l /var/log/snort -i enp0s3 -A console -c /etc/snort/snort.conf
[sudo] password for jay:
11/22-14:38:52.256726 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.11 -> 172.20.10.12
11/22-14:38:52.268500 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.12 -> 172.20.10.11
11/22-14:38:53.259981 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.11 -> 172.20.10.12
11/22-14:38:53.260492 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.12 -> 172.20.10.11
11/22-14:38:54.261716 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.11 -> 172.20.10.12
11/22-14:38:54.261859 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.12 -> 172.20.10.11
11/22-14:38:55.356279 [**] [1:99991:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 172.20.10.11 -> 172.20.10.12
```

```
Administrator: Windows PowerShell
03/29-23:45:35.831394 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:35.953554 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.091926 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.205018 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.334136 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.468595 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.576322 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.699995 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:36.878769 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.014820 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.125525 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.253099 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.443968 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.564243 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.685833 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
03/29-23:45:37.798081 [**] [1:19559:7] INDICATOR-SCAN SSH brute force login attempt [**] [Classification: Misc activity] [P
.168.1.143:22
```

## Result

Thus, Snort IDS/IPS was successfully used to detect and prevent real-time network threats using the TryHackMe platform.

The experiment enhanced understanding of intrusion detection, prevention mechanisms, and real-time network security monitoring.

### ✔ Important Note

This experiment was conducted strictly for **educational and ethical purposes** using the TryHackMe learning platform.